COMSATS University Islamabad, Lahore Campus

Terminal Examination – Spring 2025

Course Title:	Web Technologies			Course Code:	CSC336	Credit	4(3,1)
Course Instructor/s:	Asif Shahzad			Programme Name:	BSCS		
Semester:	6	Batch:	FA22	Section:	G1, G2	Date:	Jun. 28, 2025
Time Allowed:	**3 Hours**			**Maximum Marks:**	**50**		
Student's Name:				Reg. No.			
Note: Please answer only what is asked. Write very neat, keeping the font-size small. No marks for writing stuff that is not asked. You can skip the includes, middle-wares, html, head and body tags etc.							

Question 1 [4 X 5 = 20 Marks] [CLO-1] Bloom Taxonomy Level: Understanding

a) What is Service-Oriented Architecture (SOA) and Server-Side Rendering (SSR) in web applications?
b) What are micro-services in SOA? Use a labeled diagram to compare micro-services-based SOA and monolithic SOA. Explain key differences briefly.
c) What is the benefit of the PRG (Post-Redirect-Get) pattern in form handling? Explain using a request/response flow diagram. Mention the HTTP status code and header used in the redirect.
d) What is middleware in web applications? How can middleware be used to measure total request processing time on the server?

Question 2 [15 Marks] [CLO-2] Bloom Taxonomy Level: Applying

Assume an end point *example.com/book* return a book object in JSON format. The endpoint takes book-id as query string. Create an HTML form having book title (text field), published (yes/no radio button), genre (drop down list). On page load, fetch book data for Book ID 100 using AJAX and fill the form correctly.

Question 3 [3 X 5 = 15 Marks] [CLO-3] Bloom Taxonomy Level: Applying

a) The session's **`data`** array holds employees (*id, name, gender, city*). Create a handler and EJS view that lists only male employees in a table.
b) Write a GET request handler for **`/save-data`** that saves all query parameters as cookies in the browser.
c) Write a handler for an HTTP POST request that has multiple parameters e.g. **`/process`**, that shall redirect the request to the **`/show-data`** endpoint. The **`/show-data`** endpoint should then send all the parameters (which were submitted to /process) back to the browser in **JSON** format.